Module 5

Components of IoT Architecture

1) Sensors and actuators
2) Gateway
3) Backend Services

These are three components communicate via a wide variety of interfaces and protocols and enable the functionality of an IoT devices

Sensors and actuators

- These gather specific data for analysis, needing the right accuracy and frequency
- application specific and ensure that the data collected meets the accuracy

Gateway

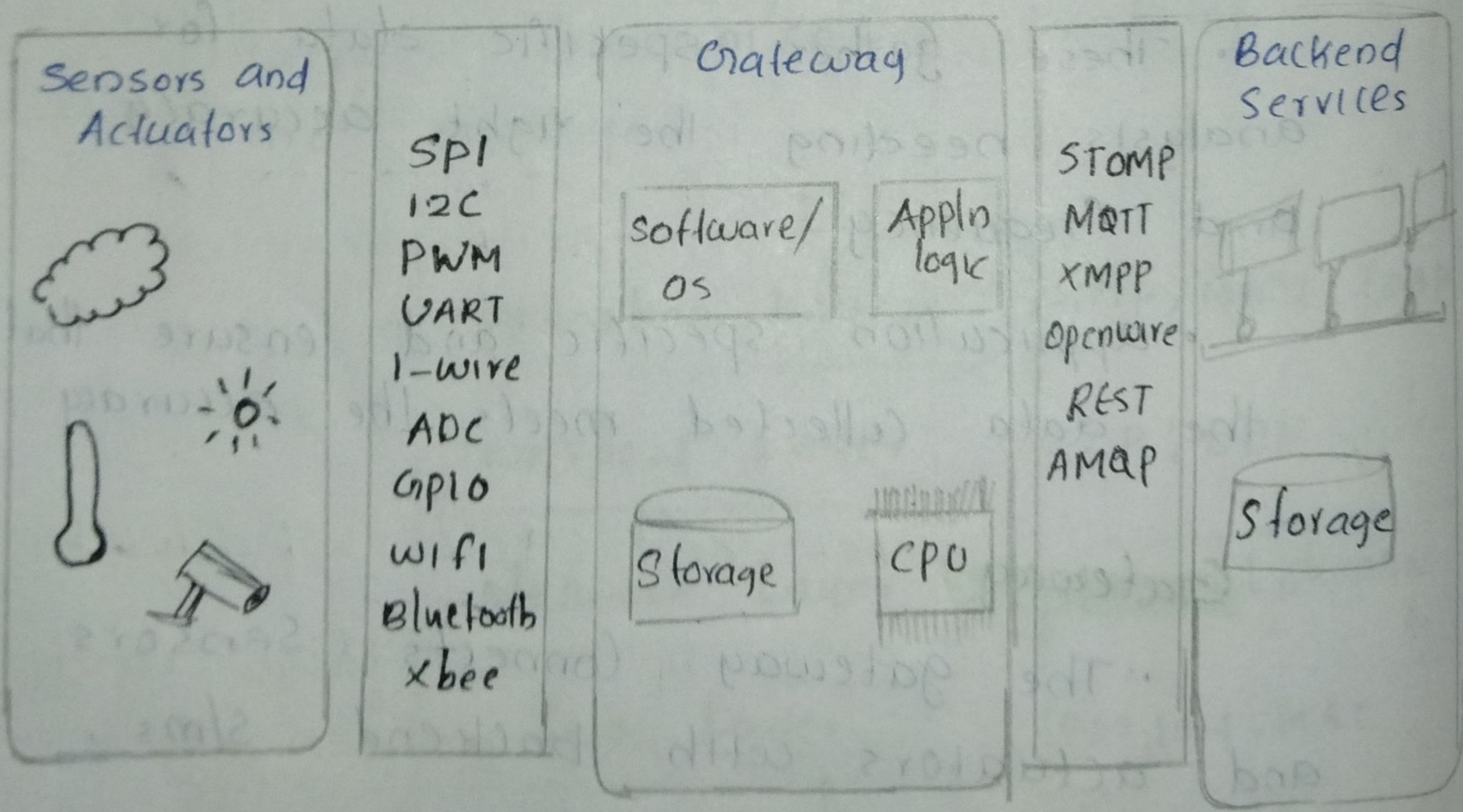
- The gateway connects sensors and actuators with backend s/ms, acting like a translator.
- It adds Internet capability to sensors and can store data if the

Internet is unreliable

Backend Services

- Used to store the IoT device data
- It can include additional functionality such as individual device configuration and analysis algorithms

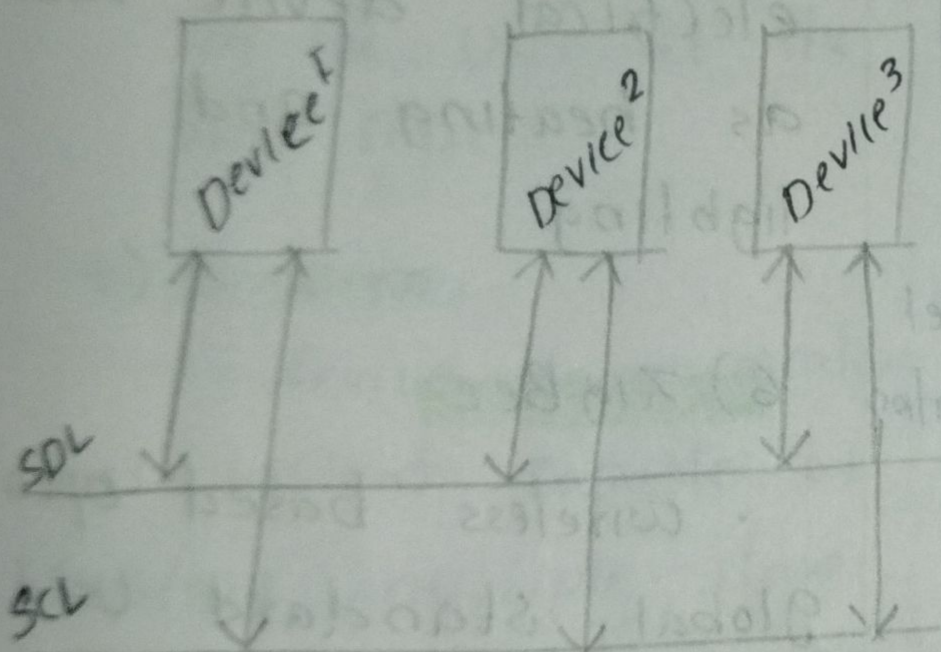
All three components need to communicate to move data to the internet and allow updates to the gateway



★ Wired Gateway

1) I²C (Inter Integrated circuit)

- widely Used for short distance communication
- It Uses 2 bidirectional lines for data communication
  1) Serial data line
  2) Serial clock line



Advantages
- Cost efficient
- Complexity is reduced

Disadvantages
- slower speed
- Half duplex communication

Wireless Gateway

1) Bluetooth Low energy
- similar to classic bluetooth
- low power consumption and cost
- larger battery life
- master/slave topology

2) 6LowPAN
- IPV6 Over Low Power wireless personal Area Networks
- allows IPV6 Packets to be sent and received within small link layer frames
- supports up to 100 nodes per network

3) INSTEON
- home automation technology that enables electrical devices to interoperate through power cables, radio frequency communication or both using a dual-mesh n/wing topology

2) Serial peripheral interface (SPI)

- It's a serial bus capable of hosting a single master with multiple slave devices per bus
- lower power consumption than I^2C, but requires more connections.

3) Universal Asynchronous Receiver/Transmitter (UART)

- refers to the h/w that converts parallel to serial communication

4) Analog to Digital converter (ADC)

- Converts a continous analog voltage to a digital number
- number of bits used to store this number relates to the accuracy of the ADC

4) Infrared

Useful for interacting with devices, such as heating controllers, air conditioners and television receivers

5) Zwave

wireless based home automation system for controlling electrical device such as heating and lighting

6) ZigBee

- wireless based open, global standard used for personal area networks.
- Use mesh networking
- Battery life lasting several years

5) Controller Area n/w (CAN)

- message based protocol
- allows multiple master device communication

6) General purpose Input output (GPIO)

- general purpose pin whose behaviour can be controlled by the user at the run time.

7) 1-wire

- device communication bus system capable of hosting multiple slave device connected to one single master

8) x10

- Protocol for communication b/w automated home-electronic devices.

7) xbee

- Is a communication module product that provides wireless end point connectivity to devices using IEEE 802.15.4 protocol
- partially compatible with zigbee
- offers fast point-to-multipoint or peer to peer communication

* **Sensors required to build the environmental sensing IOT gateway device for weather monitoring**

• Sensors have a reasonable accuracy for an external weather station, and will all be connected simultaneously

→ The wind vane has eight magnetic reed switches arranged in a dial each reed switch is connected to a different sized resistor

As the vane changes direction a magnet moves, making and breaking the switches, and therefore changing the resistance of the circuit.

→ The three cup hemispherical anemometer

Used to calculate the wind speed. As the anemometer cups rotate, a single magnet on the device closes a reed switch. completing the circuit.

→ The SHT15 temperature & humidity sensor

Very reliable low cost sensor, with accuracy of ± 0.3°C.

Gateway

- The gateway reads data from the sensors electronic interface and transmits it to a destination across the internet
- The gateway must have the appropriate electronic interface,
  - a processor with memory, either wired or wireless internet capability
  - small microprocessor such as those used by Arduino would be perfect.

Requirements used to select gateway h/w

→ Electronic Interface

→ Data persistence

→ wired or wireless internet stack

→ Data encryption

→ Data processing capability

→ programmability

→ Device security

→ future proof.

Gateway hardware

- we can divide the available h/w into groups and select the preferable h/w for the gateway based on the previously outlined requirements.

- A **Micro processor** usually only offers processing power and requires RAM and storage to be added as separate chips on PCB.

- A **Micro-Controller** has most of the basics embedded on a single chip. It is for this reason that ~~con~~ micro controllers are considered for the gateway hardware.

- more powerful micro controllers exist and are often refferd as **System on a Chip (SoC)**

- SoC usually supports an OS such as Linux or windows.

- Both micro controller and SoC need to be assembled on a PCB with other components

Gateway Software

- The more powerful a microcontroller or SoC, the more complicated the software becomes

- .NetMF platform is open source and supports high-level programming languages such as C# and Visual Basic

- .NetMF would be a good contender for the weather station gateway

- Mbed platform - supports SSL, good selection of supported h/w, C language support.
- designed specifically for IoT

- Microsoft windows 10 IOT Platform is a full OS that is targeted toward IoT devices

- There are many variant of Linux available for embedded System